Smart Complaint Management System

Software Requirements Specification — Lab 1

Name: Kanchhit | PES University | SRN: PES1UG24CS213

1. User Stories

ID	Role	User Story	Priority	Acceptance Criteria
US-01	Citizen	As a citizen, I want to submit a complaint with a photo and GPS location so that the concerned department knows exactly what and where the issue is.	Must	Complaint saved with image and map pin; confirmation ID shown.
US-02	Citizen	As a citizen, I want to track the real-time status of my complaint so that I know what stage it is in without calling a helpline.	Must	Timeline view shows status transitions with timestamps; push notification on each change.
US-03	Citizen	As a citizen, I want to rate the resolution and leave feedback so that service quality can be measured.	Must	1-5 star rating and comment saved; CSAT score computed.
US-04	Citizen	As a citizen, I want to receive SMS and email updates so that I stay informed even without the app.	Should	SMS sent on status changes for opted-in users; email with complaint summary.
US-05	Agent	As an agent, I want to see complaints routed to my department sorted by SLA urgency so that I handle the most critical ones first.	Must	Dashboard shows SLA countdown; overdue items highlighted at the top.
US-06	Agent	As an agent, I want to add internal notes and attach evidence to a complaint so that the resolution trail is documented.	Must	Notes saved with agent name and timestamp; attachments support PDF and images up to 10 MB.
US-07	Agent	As an agent, I want the system to suggest a category automatically so that I spend less time on manual triage.	Should	NLP model suggests top-3 categories with confidence scores above 70%.
US-08	Supervisor	As a supervisor, I want to escalate a complaint to another team so that stalled tickets get expert attention.	Must	Escalation transfers ticket to priority queue with a new SLA; all parties notified.
US-09	Supervisor	As a supervisor, I want a dashboard of KPIs (open, resolved, overdue, avg time) so that I can monitor team performance.	Must	Dashboard refreshes every 60 s; export to CSV available.
US-10	Admin	As an administrator, I want to configure SLA rules per category and severity so that response expectations match organisational policy.	Must	SLA rules editable via form; changes effective on new complaints immediately.

2. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow citizens to submit complaints via web or mobile, including title (max 120 chars), description (max 2000 chars), category, severity, up to 5 images (max 5 MB each), and optional GPS coordinates.	High	Pass: Complaint saved with all fields. Fail: Mandatory field missing or oversized attachment accepted.	Core submission workflow
FR-002	Functional	The system shall use an NLP model to suggest top-3 complaint categories with confidence scores; citizen may accept or override.	High	Pass: Suggestions shown with >= 70% confidence. Fail: No suggestion or wrong category ranked first.	Reduces manual triage effort
FR-003	Functional	The system shall generate a unique ticket ID (SCMS-YYYYMMDD-NNNNN) on successful submission.	High	Pass: Unique ID shown on confirmation. Fail: Duplicate ID or missing.	Tracking reference for all parties
FR-004	Functional	The system shall route new complaints to the appropriate department and agent based on category, location, and agent workload.	High	Pass: Complaint lands in correct dept queue. Fail: Routed to wrong department.	Eliminates manual assignment
FR-005	Functional	The system shall enforce a complaint lifecycle: New -> Assigned -> In Progress -> Pending Info -> Resolved -> Closed, logging every transition with timestamps.	High	Pass: Status history shows all transitions. Fail: Transition skipped or unlogged.	Auditability and transparency
FR-006	Functional	The system shall compute SLA deadlines per complaint based on category and severity rules, and trigger alerts when approaching (< 2 h) or breached.	High	Pass: Alerts sent at warning and breach thresholds. Fail: No alert on SLA breach.	Accountability for resolution times
FR-007	Functional	The system shall send notifications via in-app, email, and SMS (opt-in) on status changes, SLA warnings, and resolution confirmation.	Medium	Pass: Notification received within 60 s. Fail: Notification missing or delayed > 5 min.	Keeps citizens informed
FR-008	Functional	Upon resolution, the system shall prompt the citizen for a 1-5 star rating and optional comment within 7 days; unrated complaints auto-close.	Medium	Pass: Rating saved and CSAT updated. Fail: Rating not recorded.	Measures service quality
FR-009	Functional	The system shall allow supervisors to escalate complaints, resetting SLA timers and moving tickets to a priority queue.	High	Pass: Ticket moves to priority queue with new SLA. Fail: SLA not reset.	Handles stalled complaints
FR-010	Functional	The system shall support CRUD on user accounts with RBAC for four roles: Citizen, Agent, Supervisor, Admin.	High	Pass: Permissions enforced per role on every API call. Fail: Unauthorised access succeeds.	Security and access control

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-011	Functional	The system shall provide real-time dashboards showing open/closed counts, avg resolution time, SLA compliance rate, and CSAT trends.	Medium	Pass: Dashboard refreshes every 60 s with accurate data. Fail: Stale metrics.	Performance monitoring
FR-012	Functional	The system shall maintain an immutable audit log of every complaint action with actor ID and timestamp.	High	Pass: Full action history retrievable. Fail: Any action unlogged.	Regulatory compliance
FR-013	Functional	The system shall allow an assigned agent to resolve a complaint by recording the action taken and marking it Resolved, or to set it to Pending Info and query the citizen when details are missing.	High	Pass: Status changes to Resolved with resolution notes and citizen is notified. Fail: Complaint resolved without notes or citizen not notified.	Core resolution workflow
FR-014	Functional	The system shall allow agents to add timestamped internal notes and attach evidence files (PDF, JPEG, PNG; max 10 MB each) to an assigned complaint.	Medium	Pass: Note and file saved with agent name and timestamp. Fail: File over 10 MB accepted or note saved without author.	Documented resolution trail
NFR-001	Nonfunc.	The system shall handle 500 concurrent users with API response times under 500 ms (p99) and page loads under 2 seconds (p95).	High	Pass: Load test confirms metrics. Fail: Latency exceeds thresholds.	Responsive user experience
NFR-002	Nonfunc.	All data in transit shall use TLS 1.3; data at rest encrypted with AES-256. OWASP Top 10 mitigated.	High	Pass: Security audit confirms compliance. Fail: Any PII found unencrypted.	Data protection and privacy
NFR-003	Nonfunc.	The system shall maintain 99.9% uptime (< 8.76 h downtime/year). RPO: 1 hour, RTO: 30 minutes.	High	Pass: Uptime report confirms target. Fail: Downtime exceeds threshold.	Service reliability

3. Test Cases

Test cases follow the template: Test Case ID, Name of Module, Test case description, Pre-conditions, Test Steps, Test data, Expected Results, Actual Result, Test Result.

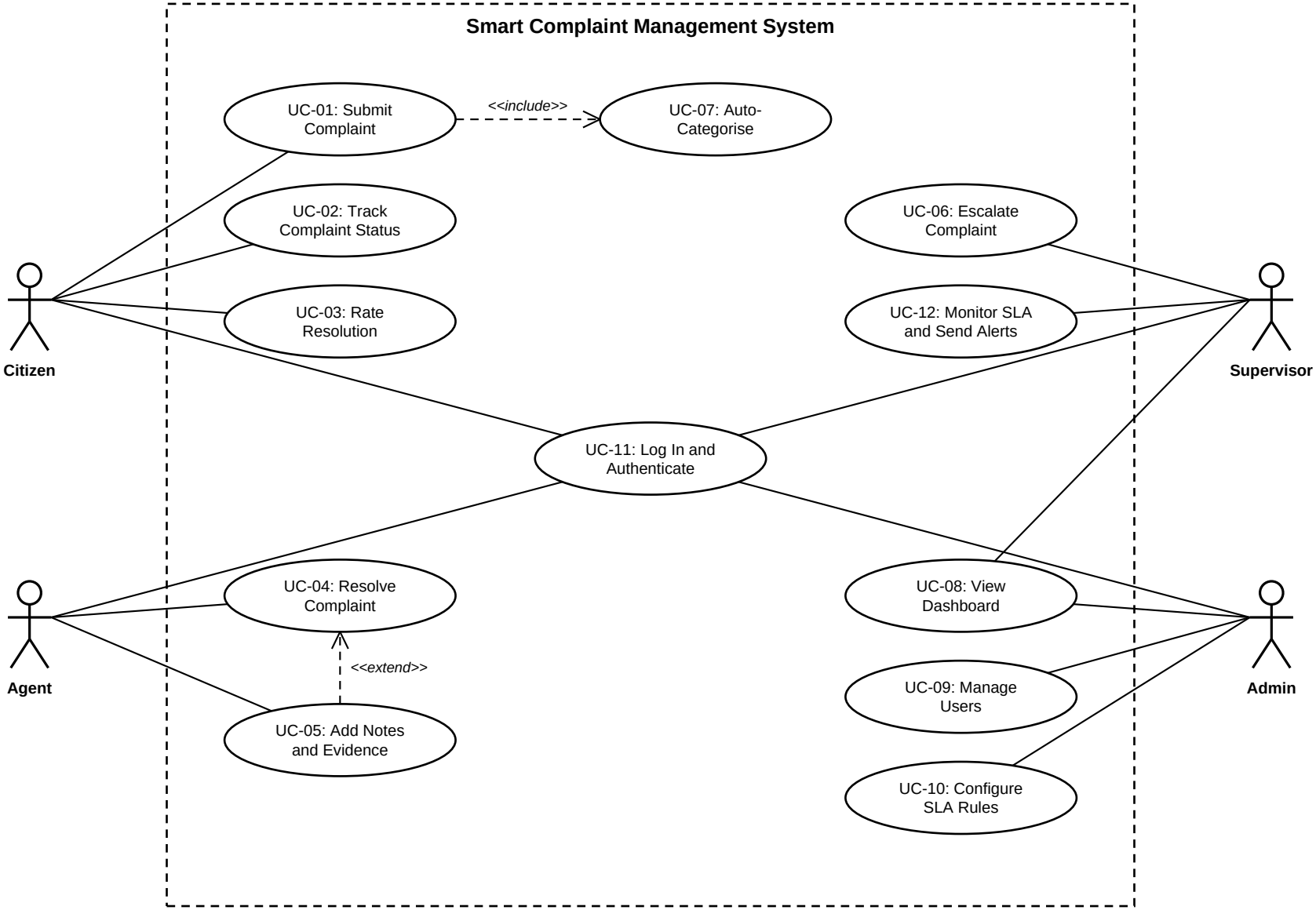
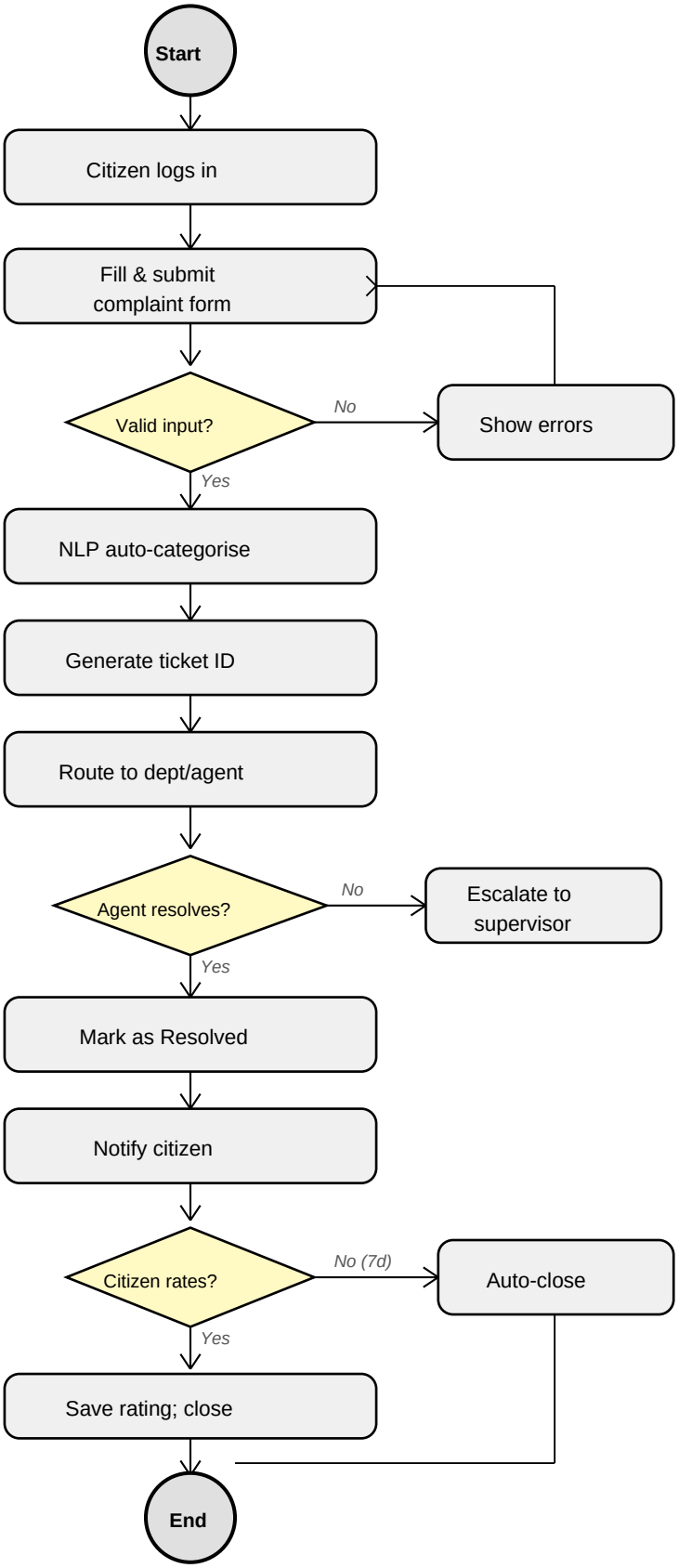
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	Complaint Submission	To test the complaint form submission with valid inputs	Citizen is logged in; on home screen	1: Click 'New Complaint' 2: Enter title 'Pothole on MG Road' 3: Enter description 4: Select category 'Roads' 5: Set severity 'High' 6: Attach image (2 MB) 7: Enable GPS 8: Click Submit	Title: Pothole on MG Road Category: Roads Severity: High Image: pothole.jpg (2 MB)	Complaint created with ticket ID SCMS-YYYYMMDD-NNNNN; confirmation screen with timeline link	Complaint created successfully with ID SCMS-20260902-00001; confirmation displayed	Pass
UT-02	Complaint Submission	To test form validation when mandatory fields are empty	Citizen is logged in	1: Click 'New Complaint' 2: Leave title empty 3: Leave category unselected 4: Click Submit	Title: (empty) Category: (none) Severity: (none)	Validation error: 'Title is required', 'Category is required' shown; form not submitted	Validation errors displayed correctly; form not submitted	Pass
UT-03	Complaint Submission	To test rejection of oversized attachment	Citizen is logged in; filling complaint form	1: Click 'Attach Image' 2: Select a 12 MB file 3: Observe response	Image: largefile.jpg (12 MB)	Error message: 'File exceeds 5 MB limit'; file not attached	Error message displayed; file rejected	Pass
IT-01	Auto-Categorisation	To test NLP model suggests correct category for complaint text	NLP model deployed and active	1: Citizen types 'water leakage from main pipe on 5th cross' 2: System processes text 3: Observe suggested categories	Description: 'water leakage from main pipe on 5th cross'	Top suggestion: 'Water Supply' with confidence >= 70%; top-3 list shown	Top suggestion: Water Supply (87%); list shown correctly	Pass
IT-02	Smart Routing	To test complaint routes to correct department and agent	Complaint submitted under category 'Water Supply' in Zone 3; agents active	1: New complaint created 2: System evaluates routing rules 3: Observe assignment	Category: Water Supply Location: Zone 3	Complaint assigned to Water Dept - Zone 3 agent with lowest current workload	Assigned to Agent Ravi (Water Dept - Zone 3, 4 active tickets)	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-03	SLA Engine	To test SLA breach triggers alert notifications	Complaint with 24 h SLA created; 22 hours elapsed	1: Wait for SLA warning threshold (2 h remaining) 2: Check agent notifications 3: Wait for SLA breach (0 h) 4: Check supervisor notifications	SLA: 24 h Elapsed: 22 h	Amber alert to agent at 2 h remaining; red alert to supervisor at breach	Amber alert sent to agent; red alert sent to supervisor at breach	Pass
ST-01	Complaint Lifecycle	To test full complaint lifecycle from submission to closure	Citizen and Agent are logged in; system active	1: Citizen submits complaint 2: System assigns to agent 3: Agent changes status to 'In Progress' 4: Agent adds resolution note 5: Agent marks 'Resolved' 6: Citizen rates 4 stars 7: Complaint auto-closes	Valid complaint data; rating: 4 stars; comment: 'Fixed well'	Status transitions: New -> Assigned -> In Progress -> Resolved -> Closed; all logged with timestamps	All status transitions logged correctly; CSAT updated	Pass
ST-02	Escalation	To test supervisor escalation of stalled complaint	Complaint in 'In Progress' for > 48 h; supervisor logged in	1: Supervisor opens overdue tab 2: Selects stalled complaint 3: Clicks 'Escalate' 4: Enters reason 5: Selects senior team 6: Confirms	Complaint ID: SCMS-20260830-00042 Reason: 'No response from field team'	Ticket moved to priority queue; new SLA timer starts; original agent and citizen notified	Ticket escalated; new SLA set; notifications sent to all parties	Pass
ST-03	User Management (RBAC)	To test admin creates agent account with correct role permissions	Admin is logged in	1: Navigate to User Management 2: Click 'Add User' 3: Enter name, email, department 4: Select role 'Agent' 5: Click Save	Name: Priya S Email: priya@scms.in Dept: Roads Role: Agent	Account created; welcome email sent; agent can log in and see only assigned complaints	Account created; welcome email sent; agent logged in with correct permissions	Pass
ST-04	Security	To test SQL injection prevention on complaint title field	System running; citizen logged in	1: In complaint title, enter: ' OR 1=1 -- 2: Fill remaining fields normally 3: Click Submit	Title: ' OR 1=1 -- Other fields: valid data	Input sanitised; no SQL execution; complaint saved with escaped characters	Input sanitised; complaint saved normally with escaped text	Pass

4. UML Use-Case Diagram

Actors: Citizen, Agent (left) — Supervisor, Admin (right)

Relationships: UC-01 (Submit Complaint) <<include>> UC-07 (Auto-Categorise); UC-05 (Add Notes and Evidence) <<extend>> UC-04 (Resolve Complaint)



4b. Workflow Diagram: Submit & Resolve Complaint